Review Article

Recent Advances in Synthesis, Characterization, and Applications of Schiff Base Ligands: A Comprehensive Review

Milad Mahjoobizadeh* | **Vahidreza Darugar** | **Fatemeh Khodabakhshi**

Department of Chemistry, University of Torbat-e Jam, Torbat-e Jam, Iran

Use your device to scan and read the article online

Citation M. Mahjoobizadeh, V. Darugar, F. Khodabakhshi, **Recent Advances in Synthesis, Characterization, and Applications of Schiff Base Ligands: A Comprehensive Review.** *Eurasian J. Sci. Technol.*, **2026**, 6(4), 352-373.

https://doi.org/10.48309/ejst.2026.588264.1334

CC BY

Article info:
Received: 2026-09-15
Accepted: 2026-11-10
Available Online: 2026-11-12
ID: EJST-2609-1334

Keywords:
Schiff base ligands; Spectroscopic techniques; Green chemistry; Biological applications; Catalysis; Materials science

ABSTRACT

Schiff base ligands have attracted considerable attention in modern chemistry due to their facile synthesis, structural diversity, and wide range of applications in coordination chemistry, medicinal chemistry, catalysis, and materials science. These compounds are commonly prepared through condensation reactions between primary amines and carbonyl compounds, producing the characteristic azomethine (–C=N–) functional group responsible for their versatile chemical behavior. This review provides a comprehensive overview of recent advances in synthesis and characterization of new Schiff base ligands. Different synthetic approaches, including conventional reflux methods, green synthesis strategies, microwave-assisted techniques, and solvent-free procedures, are discussed with emphasis on their advantages and limitations. Furthermore, important characterization techniques such as FTIR, NMR, UV–Vis spectroscopy, mass spectrometry, X-ray diffraction, and thermal analysis are summarized to highlight their role in structural confirmation and property evaluation. The biological, catalytic, environmental, and material applications of Schiff base ligands and their metal complexes are also reviewed. Finally, current challenges and future perspectives regarding sustainable synthesis, advanced functional materials, and biomedical applications are discussed. This review highlights the continuing importance of Schiff base chemistry and its potential contribution to the development of multifunctional compounds for future technological and scientific applications. The review also briefly highlights the emerging electrochemical and bioanalytical applications of Schiff base ligands, which represent a growing area of interest.

Introduction

Schiff base ligands constitute one of the most significant classes of organic compounds in modern chemistry due to their straightforward synthesis, structural versatility, and broad spectrum of applications. These compounds are generally synthesized through the condensation reaction between a primary amine and an aldehyde or ketone, leading to the formation of the characteristic azomethine (–C=N–) functional group. Since the pioneering work of Hugo Schiff in 1864, Schiff base chemistry has evolved into

*Corresponding Author: Milad Mahjoobizadeh; Email: milad_mahjoobizadeh@tjamcaas.ac.ir

an active research field encompassing organic synthesis, coordination chemistry, medicinal chemistry, analytical chemistry, and materials science [1-4].

The increasing interest in Schiff base ligands is mainly attributed to their remarkable coordination ability toward transition and lanthanide metal ions. The presence of donor atoms such as nitrogen, oxygen, and sulfur enables these ligands to form stable mono-, bi-, tri-, or polydentate complexes with various metal ions. Such complexes often exhibit enhanced physicochemical properties compared with the free ligands, including improved thermal stability, catalytic efficiency, electronic characteristics, and biological activity. Consequently, Schiff base ligands have become indispensable building blocks for the design of functional coordination compounds [5–8].

In recent years, remarkable progress has been achieved in the synthesis of new Schiff base ligands due to the development of environmentally friendly and high-efficiency synthetic methodologies. While conventional reflux synthesis remains one of the most frequently employed methods bedue tocause of its simplicity and reproducibility, alternative approaches including microwave-assisted synthesis, ultrasound-assisted synthesis, solvent-free reactions, mechanochemical methods, and green solvent systems have attracted increasing attention. These modern techniques not only reduce reaction time and energy consumption, but also improve product yield and minimize hazardous waste, supporting the principles of sustainable chemistry [9–13].

Following synthesis, comprehensive characterization is essential to confirm the molecular structure and evaluate the physicochemical properties of newly prepared Schiff base ligands. Fourier Transform Infrared Spectroscopy (FTIR) is routinely employed to verify the formation of the azomethine linkage, whereas Nuclear Magnetic Resonance (^{1}H- and ^{13}C-NMR) spectroscopy provides detailed structural information regarding the ligand framework. Additional techniques such as UV–Visible spectroscopy, Mass Spectrometry (MS), Elemental Analysis (CHN), X-ray Diffraction (XRD), Thermogravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC), Scanning Electron Microscopy (SEM), and Transmission Electron Microscopy (TEM) collectively contribute to the complete characterization of these compounds. The combined use of complementary analytical methods significantly increases the reliability of structural identification and facilitates the correlation between molecular structure and functional properties [14–17].

Beyond their structural importance, Schiff base ligands and their metal complexes have demonstrated considerable potential in numerous scientific and industrial applications. Extensive investigations have revealed significant antibacterial, antifungal, antiviral, antioxidant, anti-inflammatory, and anticancer activities, making these compounds attractive candidates for medicinal chemistry and drug development. Moreover, Schiff base derivatives have been successfully applied in homogeneous and heterogeneous catalysis, corrosion inhibition, fluorescent sensing, environmental remediation, heavy metal adsorption, and advanced functional materials. Their tunable molecular architecture enables continuous structural modification, allowing researchers to optimize their performance for specific applications [18–20].

Although several reviews on Schiff bases have been published in recent years, most of them focus on a specific application area or a particular class of ligands. For example, recent reviews have concentrated on the anticancer properties of tetradentate N_2O_2 Schiff base complexes, the biological activity of indole-derived Schiff bases, or the coordination chemistry of salicylaldehyde-derived penta-, hexa-, and heptadentate ligands. Other reviews have discussed the computational chemistry and biological activity of Schiff base metal complexes, or the environmentally friendly synthesis of Schiff base ligands. However, these works are generally limited to a single dimension—specific ligands, specific applications, or specific methods—and lack an integrated analysis linking synthetic methods, characterization strategies, and multifunctional applications [21,22].

The unique contribution of the present review lies in three aspects: (1) it provides a quantitative and critical comparison of

conventional reflux, green synthesis, microwave-assisted, and solvent-free methods in terms of reaction time, yield, solvent requirements, energy consumption, reproducibility, and scalability; (2) it correlates characterization techniques with their specific roles in structural confirmation and property evaluation; and (3) it spans biological, catalytic, environmental, and materials science applications, revealing how structural modification influences multifunctional performance. Notably, no review published in 2025–2026 has adopted such an integrated perspective combining critical comparison of synthetic methodologies with cross-disciplinary application analysis. Although the primary focus of this review is on synthesis, characterization, and structural/spectroscopic perspectives, emerging electrochemical and bioanalytical applications are also briefly discussed to provide a broader perspective on the versatility of Schiff base ligands [23-25].

The rapid expansion of research on Schiff base ligands during the last decade has generated a substantial body of literature. Nevertheless, the increasing diversity of reported ligands and analytical approaches highlights the need for an updated and comprehensive review integrating these developments. Therefore, the present review summarizes recent advances in the synthesis and characterization of new Schiff base ligands, critically discusses conventional and green synthetic methodologies, describes the principal analytical techniques used for structural confirmation, and highlights the most important biological, catalytic, and industrial applications. Furthermore, current challenges and future research perspectives are presented to provide a comprehensive understanding of this continuously evolving field [26-28].

Literature Search Strategy

This review was conducted following a systematic approach to ensure comprehensive coverage of recent advances in Schiff base ligands. The methodology was adapted from the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, where applicable.

Limitations of the Search

This review focused primarily on English-language publications and may have excluded relevant studies published in other languages. Additionally, the rapid growth of Schiff base research means that some very recent studies may not have been included. Nevertheless, the selected references represent a comprehensive and up-to-date overview of the field [29,30].

Chemistry of Schiff Base Ligands

Schiff base ligands are a distinctive class of organic compounds characterized by the presence of the azomethine or imine functional group (–C=N–), which is formed through the condensation reaction between a primary amine and a carbonyl compound, typically an aldehyde or ketone. The formation of this imine linkage is generally accompanied by the elimination of one molecule of water and is influenced by several factors, including the nature of the reactants, solvent, temperature, catalyst, and reaction time. Due to the simplicity of this reaction and the vast variety of commercially available amines and carbonyl compounds, an enormous number of Schiff base ligands with diverse molecular architectures have been synthesized and investigated [31-33].

The electronic structure of the azomethine group plays a crucial role in determining the chemical behavior of Schiff base ligands. The lone pair of electrons on the nitrogen atom, together with adjacent donor atoms such as oxygen, sulfur, or additional nitrogen atoms, provides excellent coordination sites for metal ions. Consequently, Schiff base ligands may function as mono-, bi-, tri-, tetra-, or polydentate ligands depending on their molecular structure. The flexibility of these donor atoms allows the formation of stable coordination complexes with transition metals, lanthanides, and even some main-group elements, making Schiff base ligands indispensable in coordination chemistry [34,35].

One of the major advantages of Schiff base chemistry is the remarkable structural diversity that can be achieved through simple modification of the starting materials. Aromatic aldehydes, heterocyclic aldehydes, substituted

ketones, and various aliphatic or aromatic amines can be combined to generate ligands possessing different steric and electronic properties. The incorporation of electron-donating or electron-withdrawing substituents significantly influences the electron density around the azomethine group, thereby affecting metal-binding affinity, stability, solubility, and biological activity. Such structural tunability enables the rational design of ligands for specific catalytic, medicinal, or environmental applications [10,11].

Schiff base ligands may exist in different tautomeric forms depending on their molecular framework and surrounding conditions. Keto–enol and imine–enamine tautomerism are frequently observed in ligands containing hydroxyl or other proton-donating functional groups. These tautomeric equilibria influence the coordination behavior, spectroscopic properties, and overall stability of the ligands. Additionally, geometric isomerism around the C=N bond may result in E and Z configurations, with the E isomer generally being more thermodynamically stable due to the reduced steric repulsion. Understanding these structural features is essential for interpreting spectroscopic data and predicting the coordination mode of newly synthesized Schiff base ligands [12,13].

The physicochemical properties of Schiff base ligands are strongly dependent on their molecular structures. Parameters such as polarity, molecular planarity, hydrogen-bonding capability, and conjugation influence their solubility, thermal stability, optical behavior, and reactivity. Extended π-conjugation frequently enhances electronic delocalization, leading to improved optical and electrochemical properties. Likewise, intramolecular hydrogen bonding may stabilize the molecular framework and affect both crystal packing and metal coordination. These characteristics explain the increasing use of Schiff base ligands in advanced functional materials, sensing devices, and optoelectronic applications (Figure 1) [14,15].

Overall, the combination of facile synthesis, structural flexibility, tunable electronic properties, and excellent coordination ability has established Schiff base ligands as one of the most versatile ligand families in contemporary chemistry. Their unique chemical characteristics provide the fundamental basis for the development of new synthetic methodologies, advanced characterization techniques, and diverse practical applications, which are discussed in the following sections [16-18].

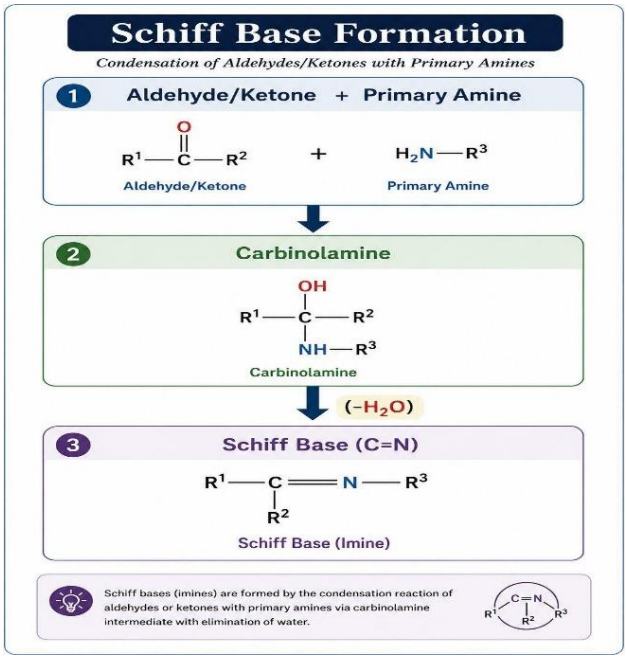


Figure 1: General mechanism for the formation of Schiff base ligands [15]

Synthesis of Schiff Base Ligands

Conventional Synthesis

The conventional synthesis of Schiff base ligands is the most widely employed and historically established approach for the preparation of these compounds. In this method, a primary amine reacts with an aldehyde or ketone in an appropriate solvent, most commonly ethanol, methanol, or other polar organic solvents, under reflux conditions. The reaction proceeds through nucleophilic attack of the amino group on the carbonyl carbon, followed by the formation of a carbinolamine intermediate. Subsequent dehydration leads to the formation of the characteristic azomethine (–C=N–) linkage, producing the desired Schiff base ligand. The reaction mechanism is relatively simple; however, several experimental parameters—including temperature, reaction time, solvent polarity, catalyst type, and the electronic nature of the substituents—significantly influence the reaction efficiency and product yield [19,20].

Conventional reflux synthesis remains popular because it requires readily available laboratory equipment, provides reproducible reaction conditions, and is applicable to a broad range of aldehydes, ketones, and primary amines. In many reported studies, reaction times vary from one to several hours depending on the reactivity of the starting materials. Although satisfactory yields are frequently obtained, prolonged heating may increase energy consumption and occasionally promote undesired side reactions or by-product formation. Therefore, careful optimization of reaction conditions is often required to maximize product purity and yield [21,22].

Various catalysts have been employed to improve the efficiency of conventional Schiff base synthesis. Both acidic and basic catalysts can accelerate the condensation reaction by facilitating the formation and dehydration of the intermediate species. Additionally, molecular sieves and other dehydrating agents are sometimes introduced to remove the water produced during the reaction, thereby shifting the equilibrium toward Schiff base formation. The choice of catalyst depends on the nature of the reactants and the desired product, while solvent selection also plays a critical role in determining reaction kinetics and product crystallization [23,24].

Despite the emergence of numerous modern synthetic techniques, conventional synthesis continues to serve as the benchmark for evaluating new synthetic methodologies. Its operational simplicity, low experimental cost, and broad substrate compatibility make it particularly suitable for laboratory-scale preparation and preliminary investigations of newly designed Schiff base ligands. Nevertheless, the increasing demand for environmentally sustainable chemistry has encouraged researchers to develop greener and more energy-efficient alternatives, which are discussed in the following sections [25,26].

Figure 2 depicts formation of the ligand was previously reported by Tamizh and the group by reacting salicylaldehyde and 2-aminophenol in acetonitrile [3].

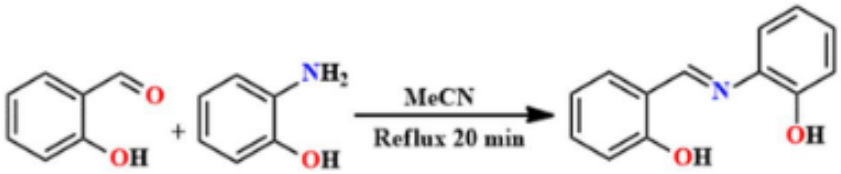


Figure 2: Synthetic protocol for Schiff base [3]

Green Synthesis

The green synthesis has emerged as an attractive alternative to conventional methods for the preparation of Schiff base ligands because it follows the principles of sustainable and environmentally friendly chemistry. This approach aims to minimize the use of hazardous chemicals, reduce energy consumption, decrease waste generation, and employ renewable or less toxic solvents whenever possible. In recent years, increasing

environmental concerns and stricter regulations have encouraged researchers to develop greener synthetic protocols without compromising reaction efficiency or product quality [27,28].

One of the most common green strategies involves replacing traditional organic solvents with environmentally benign media such as water, ethanol, ionic liquids, or deep eutectic solvents. In many cases, solvent-free conditions have also been successfully applied, eliminating the need for volatile organic compounds and simplifying product isolation. These modifications not only reduce environmental impact, but also improve atom economy and overall process sustainability. Furthermore, the use of recyclable catalysts and renewable raw materials has become an important aspect of green Schiff base synthesis [29,30].

Green synthetic approaches have been reported in many studies to offer practical advantages such as shorter reaction times, improved yields, lower energy consumption, and simpler purification. However, these benefits are not guaranteed and depend on the specific solvent system, substrate solubility, and reaction scale. In some cases, green solvents such as water or ionic liquids have led to lower yields or required longer reaction times compared with conventional organic solvents [16]. Therefore, the selection of a green synthetic route should be based on experimental optimization rather than assumed universal superiority.

Despite its numerous advantages, green synthesis also presents certain challenges. Some environmentally friendly solvents are not universally compatible with all reactants, and the optimization of reaction conditions may require additional experimental effort. Nevertheless, continuous advances in catalyst design, solvent engineering, and sustainable reaction technologies are expected to overcome these limitations. Consequently, green chemistry is anticipated to play an increasingly important role in the future development of novel Schiff base ligands with improved efficiency, safety, and environmental compatibility [31,32].

Figure 3 illustrates the rapid synthesis of seven novel bis-Schiff base compounds. These compounds were readily prepared by condensing benzidine with seven different aromatic aldehydes, as shown in Figure 3 [25].

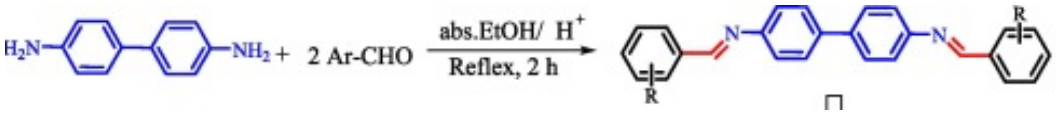


Figure 3: General reaction for the synthesis of a novel Schiff base [25]

Microwave-Assisted Synthesis

Microwave-assisted synthesis has become one of the most efficient modern techniques for the preparation of Schiff base ligands due to its ability to significantly reduce reaction time and improve energy efficiency. Unlike conventional heating methods, where heat is transferred from the external surface of the reaction vessel toward the reaction mixture, microwave irradiation provides direct molecular heating through interaction with polar molecules and ions. This unique heating mechanism can accelerate reaction kinetics and promote faster formation of the azomethine (–C=N–) linkage between carbonyl compounds and primary amines [5,9].

Several studies have demonstrated that microwave-assisted synthesis can successfully produce Schiff base ligands with high purity and excellent yields within minutes compared with the several hours often required for traditional reflux methods. The rapid energy transfer provided by microwave irradiation enhances molecular collisions and improves the efficiency of condensation reactions. Moreover, this method frequently requires smaller amounts of solvents, making it attractive from both economic and environmental perspectives [10].

Microwave-assisted procedures have been applied to the synthesis of a wide variety of Schiff base derivatives containing different functional groups and molecular architectures. The method is particularly useful for preparing ligands intended for coordination chemistry because it often provides products with improved crystallinity and reduced formation of unwanted by-products. Additionally, microwave

irradiation can be combined with green solvents or solvent-free conditions, further enhancing the sustainability of the synthetic process [11,13].

Despite these advantages, microwave-assisted synthesis requires careful optimization of experimental parameters, including microwave power, irradiation time, solvent properties, and reactant concentration. Excessive microwave energy may cause decomposition of sensitive organic molecules, while insufficient irradiation may lead to incomplete reactions. Furthermore, specialized microwave reactors may not always be available in conventional laboratories, which can limit widespread application. Nevertheless, due to its speed, efficiency, and compatibility with green chemistry principles, microwave-assisted synthesis continues to represent an important approach for the development of new Schiff base ligands [15].

Solvent-Free Synthesis

Solvent-free synthesis represents another important green approach for the preparation of Schiff base ligands, offering significant advantages in terms of environmental sustainability, operational simplicity, and reduced chemical waste. In this strategy, the condensation reaction between primary amines and aldehydes or ketones is performed without the addition of conventional organic solvents. The absence of solvents reduces the emission of volatile organic compounds (VOCs), decreases purification requirements, and improves the overall safety of the synthetic process [9].

The solvent-free approach can be carried out using various activation methods, including grinding, heating, mechanochemical techniques, and microwave irradiation. In many cases, mechanical mixing of the reactants provides sufficient energy to promote the formation of the azomethine bond, resulting in efficient synthesis with high product yields. This method is particularly attractive for laboratories aiming to reduce solvent consumption and develop more sustainable synthetic procedures [10].

Compared with conventional reflux methods, solvent-free synthesis often provides shorter reaction times, simpler reaction procedures, and easier product recovery. The direct interaction between reactant molecules can enhance reaction efficiency and improve atom economy. Furthermore, the reduced number of reaction components decreases the possibility of contamination and makes the method suitable for large-scale preparation of certain Schiff base derivatives [11].

However, solvent-free synthesis also has some limitations, including difficulties in controlling reaction temperature, mixing efficiency, and heat distribution, especially when reactions involve solid reactants or complex molecular structures. Optimization of grinding time, reactant ratio, and external energy input is therefore necessary to achieve reproducible results. Despite these challenges, solvent-free methods continue to gain importance as sustainable alternatives for the synthesis of Schiff base ligands and are expected to contribute significantly to future developments in green organic chemistry [12].

Table 1 shows that overall, the development of conventional, green, microwave-assisted, and solvent-free strategies demonstrates the remarkable adaptability of Schiff base chemistry. The selection of an appropriate synthetic route depends on the desired ligand structure, reaction efficiency, environmental considerations, and intended application of the final compound.

Table 1: Comparison of different synthetic methods for Schiff base ligands

Synthetic method	Main features	Advantages	Limitations
Conventional synthesis	Condensation reaction under reflux conditions	Simple, reliable, and widely used	Longreaction time and high solvent consumption
Green synthesis	Use of environmentally friendly solvents	Less waste, safer, and sustainable method	Optimization may be required
Assisted-Microwave synthesis	Microwave irradiation for rapid heating	Short reaction time high efficiency	Requires special equipment
Free-solvent synthesis	Reaction without conventional solvents	Reducted waste and easy purification	Control of temperature can be difficult

Critical Analysis of Synthetic Methods

A critical comparison of the four main synthetic approaches reveals that no single method is universally superior is given in Table 2. Conventional reflux remains the method of choice for reproducibility and broad substrate scope, but its environmental footprint is significant. Green synthesis offers the best sustainability profile but requires careful solvent selection and is substrate-dependent. Microwave-assisted synthesis provides the fastest reaction times and high yields, but its scalability is limited by equipment cost and uneven heating. Solvent-free methods are the most environmentally benign but suffer from poor control over reaction parameters. Therefore, the selection of a synthetic route should be guided by the target ligand structure, desired scale, available equipment, and environmental constraints. Future research should focus on hybrid approaches—such as microwave-assisted solvent-free synthesis—that combine the advantages of multiple methods while minimizing their individual limitations.

Table 2: Critical comparison of synthetic methods for Schiff base ligands

Method	Reaction Time	Yield (%)	Solvent Requirement	Energy Consumption	Reproducibility	Scalability	Key Limitation
Conventional reflux	2–12 h	60–95	High (50–100 mL)	High	Excellent	Good (batch)	Long time, high solvent waste
Green synthesis	10–60 min	70–98	Low/None	Low–Moderate	Good	Moderate	Solvent-substrate compatibility
Microwave-assisted	2–15 min	80–98	Low	Low (per reaction)	Moderate	Poor	Specialized equipment, uneven heating
Solvent-free	5–30 min	70–95	None	Low	Moderate	Limited	Poor temperature/ mixing control

Characterization Techniques

The characterization of newly synthesized Schiff base ligands is a fundamental step in confirming their molecular structures, identifying functional groups, and evaluating their physicochemical properties, as collected in Table 3. Since Schiff base compounds possess diverse structural frameworks and coordination capabilities, a combination of spectroscopic, microscopic, and analytical techniques is generally required to obtain reliable structural information. Among these techniques, FTIR, NMR, Ultraviolet–Visible (UV–Vis) spectroscopy, MS, XRD, and thermal analysis are the most commonly employed methods for structural confirmation and property evaluation [5].

FTIR

FTIR is one of the most widely used techniques for the identification and characterization of Schiff base ligands.

FTIR analysis provides valuable information regarding the functional groups present in the molecular structure and is particularly important for confirming the formation of the characteristic azomethine (–C=N–) bond. The appearance of a strong absorption band corresponding to the imine stretching vibration generally provides direct evidence for successful condensation between the amine and carbonyl precursors [6].

In Schiff base compounds, the stretching vibration of the azomethine group (C=N) is typically observed in the range of approximately 1,600–1,700 cm^{-1}, depending on the electronic nature of substituents and molecular environment. The disappearance of the characteristic carbonyl (C=O) stretching band of the aldehyde or ketone precursor and the appearance of a new imine band are commonly considered important indicators of Schiff base formation. Additionally, FTIR spectra provide information about other functional groups, including hydroxyl (O–H), amino (N–H), and thiol (S–H) groups, which may influence hydrogen bonding and metal coordination behavior [7].

FTIR spectroscopy is also extensively applied for investigating the coordination behavior of Schiff base ligands with metal ions. Upon complex formation, changes in the position and intensity of characteristic absorption bands can indicate the involvement of donor atoms such as nitrogen, oxygen, or sulfur in metal coordination. For example, shifts in the azomethine stretching frequency after complexation are frequently used as evidence for interaction between the imine nitrogen atom and the metal center. Similarly, changes in hydroxyl-related vibrations may indicate deprotonation and coordination through oxygen atoms [8,9].

Although FTIR spectroscopy alone cannot provide complete structural determination, its simplicity, rapid analysis, and ability to identify key functional groups make it an essential preliminary technique in the characterization of new Schiff base ligands. When combined with complementary analytical methods such as NMR, MS, XRD, and FTIR contributes significantly to the accurate determination of molecular structures and coordination modes [10].

Critical assessment: Although FTIR is a rapid, cost-effective, and non-destructive technique, its structural information is inherently limited. The appearance of a C=N stretching band in the 1,600–1,700 cm^{-1} region is widely used as supporting evidence for Schiff base formation, but it should not be regarded as complete structural proof. This is because several other functional groups (*e.g.*, C=C, C=O, and N–H bending and aromatic ring vibrations) absorb in the same region, leading to overlapping bands that can complicate interpretation. Furthermore, FTIR cannot provide information on molecular connectivity, stereochemistry, or the position of substituents.

Therefore, FTIR evidence must always be corroborated by complementary techniques such as ^{1}H-/^{13}C-NMR, MS, and, where possible, single-crystal XRD. For metal complexes, shifts in the C=N band may suggest coordination through the imine nitrogen, but definitive proof of the coordination mode requires XRD or other structural methods.

NMR

NMR spectroscopy is one of the most powerful analytical techniques for the structural elucidation of organic compounds, including Schiff base ligands. Unlike FTIR spectroscopy, which mainly provides information about functional groups, NMR spectroscopy offers detailed information regarding the chemical environment, connectivity, and molecular framework of synthesized compounds. Both proton NMR (^{1}H NMR) and carbon NMR (^{13}C NMR) spectroscopy are widely employed to confirm the structures of newly prepared Schiff base ligands [4].

In the ^{1}H-NMR spectra of Schiff base compounds, the characteristic signal corresponding to the azomethine proton (–CH=N–) is considered one of the most important features confirming imine formation. This signal commonly appears in the downfield region due to the deshielding effect of the imine group and its conjugated environment. The disappearance of aldehydic proton signals from the starting carbonyl compounds and the appearance of a new azomethine proton resonance provide strong evidence for successful condensation reactions. Additionally, signals associated with aromatic protons, aliphatic groups, hydroxyl groups, and other substituents provide valuable information about the overall molecular structure [5].

The ^{13}C-NMR spectroscopy provides further confirmation by identifying the carbon environments within the Schiff base framework. The carbon atom of the azomethine group generally exhibits a characteristic resonance at a lower magnetic field compared with other carbon atoms due to its electron-deficient nature. Analysis of ^{13}C-NMR data enables researchers to verify the formation of the C=N bond and evaluate the influence of substituents on the electronic properties of the ligand. Furthermore, comparison of NMR spectra before and after synthesis can provide evidence for structural transformation during the condensation process [6].

NMR spectroscopy is particularly valuable for Schiff base ligands that are diamagnetic and sufficiently soluble in common deuterated solvents. It allows the determination of molecular symmetry, tautomeric behavior, hydrogen bonding interactions, and conformational characteristics. In some cases, variable-temperature NMR studies are also employed to investigate dynamic structural changes and equilibrium between different tautomeric forms. These analyses provide deeper understanding of the relationship between ligand structure and chemical behavior [9].

Although NMR spectroscopy is mainly applied to the characterization of free Schiff base ligands, it may also provide important information about structural changes occurring during metal coordination. However, the application of conventional NMR techniques to paramagnetic metal complexes is often limited due to signal broadening and magnetic interactions. Therefore, NMR results are commonly combined with other spectroscopic and analytical techniques to achieve complete structural characterization of Schiff base compounds and their derivatives [10].

Critical assessment: NMR spectroscopy is among the most powerful tools for structural elucidation of diamagnetic Schiff base ligands, providing detailed information on chemical environment, connectivity, symmetry, and tautomeric behavior. However, its applicability is limited by several factors. First, NMR requires the compound to be soluble in deuterated solvents, which can be problematic for highly conjugated or polymeric Schiff bases. Second, paramagnetic metal complexes often produce broadened or shifted signals, making conventional NMR analysis difficult or impossible. Third, NMR cannot directly confirm the presence of a C=N bond in the absence of a reference or comparison with the starting materials. Therefore, NMR data should be interpreted alongside FTIR, MS, and XRD to establish a complete structural picture.

UV–Vis Spectroscopy

UV–Vis spectroscopy is an important analytical technique used for investigating the electronic properties of Schiff base ligands and their metal complexes. This method provides valuable information regarding electronic transitions, conjugation effects, and coordination behavior. Due to the presence of the azomethine group and

often extended aromatic systems, Schiff base compounds generally exhibit characteristic absorption bands associated with π→π* and n→π* electronic transitions. These transitions are highly dependent on molecular structure, substituent effects, and the degree of conjugation within the ligand framework [2,3].

In free Schiff base ligands, the absorption bands observed in UV–Vis spectra are mainly attributed to transitions involving the imine group and aromatic rings. The π→π* transition usually appears at shorter wavelengths and is associated with conjugated carbon-carbon and carbon-nitrogen systems, whereas the n→π* transition generally occurs at longer wavelengths due to excitation of non-bonding electrons located on heteroatoms such as nitrogen and oxygen. Changes in these absorption bands can provide useful information about electronic interactions and structural modifications within the ligand molecule [4-6].

UV–Vis spectroscopy is particularly significant in coordination chemistry because it allows the investigation of metal–ligand interactions after complex formation. The appearance of new absorption bands or shifts in existing bands after metal coordination may indicate changes in electronic structure caused by bonding between donor atoms of the Schiff base ligand and the metal center. In transition metal complexes, additional absorption features associated with d–d transitions and charge transfer processes can provide information about the electronic configuration and possible geometry of the complexes [7,8].

In addition to structural characterization, UV–Vis spectroscopy is widely used for studying the photophysical properties of Schiff base derivatives. Some Schiff base ligands exhibit fluorescence behavior due to their conjugated structures and rigid molecular frameworks, making them suitable candidates for sensing applications. The relationship between molecular structure, electronic transitions, and optical properties has therefore increased interest in Schiff base compounds for fluorescent sensors, optoelectronic materials, and analytical applications [9,10].

Although UV–Vis spectroscopy provides valuable information about electronic properties and coordination effects, it is generally used together with other characterization techniques such as FTIR, NMR, MS, and XRD. The combination of these complementary methods enables a more complete understanding of the structure, bonding behavior, and functional properties of newly synthesized Schiff base ligands and their complexes [11].

Critical assessment: UV-Vis spectroscopy is valuable for probing electronic transitions (π→π, *n*→*π*), conjugation, and metal–ligand charge transfer, and it is particularly useful for monitoring complex formation and photophysical properties. However, the technique provides low structural resolution: it cannot identify functional groups, molecular connectivity, or stereochemistry. Absorption bands are often broad and overlapping, and assignments can be ambiguous without complementary data. Moreover, UV-Vis is highly sensitive to solvent, pH, and concentration, which can complicate reproducibility. Therefore, UV-Vis should be regarded as a complementary technique for studying electronic properties rather than a primary structural tool.

MS

MS is a powerful analytical technique widely used for determining the molecular mass, molecular formula, and structural information of newly synthesized Schiff base ligands. This technique is particularly valuable because it provides direct evidence for the successful formation of the desired molecular structure by comparing the experimentally measured molecular ion peak with the calculated molecular weight of the proposed compound [2]. In the mass spectra of Schiff base ligands, the molecular ion peak corresponding to the intact molecular species is generally considered an important characteristic signal for structural confirmation. The observed mass-to-charge ratio (m/z) value allows researchers to verify the molecular weight of the synthesized ligand and assess its purity. Additionally, fragmentation patterns obtained during mass spectrometric analysis provide useful information about the stability of different

structural units and the possible cleavage pathways within the molecule [3].

Various ionization techniques, including Electron Ionization (EI), Electrospray Ionization (ESI), and Matrix-Assisted Laser Desorption/Ionization (MALDI), have been applied for the characterization of Schiff base compounds. Among these methods, ESI-MS is frequently used due to its soft ionization capability, which allows the analysis of relatively fragile organic molecules with minimal fragmentation. This feature is particularly beneficial for Schiff base ligands containing multiple functional groups or complex molecular architectures [4].

MS is also useful for investigating metal complexes derived from Schiff base ligands. The appearance of molecular ion peaks corresponding to the complex species, along with characteristic fragmentation patterns, can provide valuable information regarding metal–ligand interactions and the stability of coordination compounds. When combined with spectroscopic techniques such as FTIR, NMR, and UV–Vis spectroscopy, MS analysis contributes to a more complete understanding of the molecular structure and composition of Schiff base derivatives [5].

Despite its advantages, MS generally provides limited information about three-dimensional molecular arrangements and coordination geometry. Therefore, it is commonly used as a complementary technique alongside XRD and spectroscopic methods. Nevertheless, its high sensitivity, accuracy, and ability to confirm molecular mass make MS an essential tool in the characterization of newly synthesized Schiff base ligands [6].

Critical assessment: MS provides highly reliable information on molecular mass and formula, and its sensitivity allows detection of trace impurities. However, MS generally cannot distinguish between isomers (*e.g.*, E/Z configurations) or provide three-dimensional structural information. Fragmentation patterns can be complex and may require tandem MS (MS/MS) or high-resolution MS for unambiguous interpretation. Additionally, some Schiff base ligands are thermally labile or non-volatile, limiting the applicability of EI-MS and requiring softer ionization techniques such as ESI or MALDI. Thus, MS is best used as a confirmatory tool alongside NMR, FTIR, and XRD.

XRD Analysis

XRD is an important analytical technique used for determining the crystalline structure, phase composition, and molecular arrangement of solid Schiff base ligands and their metal complexes. Unlike spectroscopic methods that mainly provide information about functional groups and bonding characteristics, XRD analysis offers direct information about the three-dimensional arrangement of atoms within crystalline materials. Therefore, it is considered one of the most reliable techniques for confirming the structural features of crystalline Schiff base derivatives [12].

The XRD pattern of a crystalline compound is characterized by specific diffraction peaks corresponding to the arrangement of atoms within the crystal lattice. The position, intensity, and width of these diffraction peaks provide information about crystallographic parameters, crystal symmetry, phase purity, and particle size. For Schiff base ligands, XRD analysis can confirm the formation of well-defined crystalline structures and identify possible changes in crystal packing caused by different substituents or intermolecular interactions such as hydrogen bonding [13].

XRD is particularly valuable in the characterization of metal complexes containing Schiff base ligands because it can provide detailed information about coordination geometry and metal–ligand interactions. Single-crystal XRD, when suitable crystals are obtained, remains the most definitive method for determining molecular structures, including bond lengths, bond angles, coordination environments, and spatial arrangement of ligand donor atoms around the metal center. These structural parameters are essential for understanding the relationship between molecular architecture and the observed chemical or biological properties of the complexes [14].

In addition to structural determination, powder XRD analysis is frequently used for evaluating phase purity and crystallinity of synthesized

materials. This technique is especially important for Schiff base-based functional materials, nanoparticles, and coordination polymers, where crystal structure strongly influences properties such as catalytic activity, magnetic behavior, optical performance, and stability. Changes in diffraction patterns before and after chemical modification can also provide evidence for successful formation of new materials [15].

Although XRD provides highly valuable structural information, its application is generally limited to crystalline samples. Amorphous or poorly crystalline Schiff base compounds may require additional characterization methods to obtain complete structural information. Therefore, XRD results are usually interpreted together with spectroscopic techniques such as FTIR, NMR, UV–Vis, and MS to achieve a comprehensive understanding of newly synthesized Schiff base ligands and their complexes [16].

Critical assessment: Single-crystal XRD is the most definitive method for determining molecular structure, providing bond lengths, bond angles, coordination geometry, and crystal packing. However, its major limitation is the requirement for suitable single crystals, which are often difficult to obtain for Schiff base ligands and their complexes. Powder XRD can confirm crystallinity and phase purity but cannot provide atomic-level structural details. Furthermore, XRD is not suitable for amorphous or poorly crystalline samples. Therefore, XRD results are typically combined with spectroscopic and thermal methods for a comprehensive characterization.

Thermal Analysis (TGA and DSC)

Thermal analysis techniques, particularly TGA and DSC, are widely used to investigate the thermal stability, decomposition behavior, and phase transitions of Schiff base ligands and their metal complexes. These techniques provide important information regarding the resistance of synthesized compounds toward temperature changes and help determine their potential suitability for applications requiring thermal stability [15].

Thermogravimetric Analysis measures the change in sample mass as a function of temperature under controlled heating conditions. The obtained thermogravimetric curve provides information about different stages of thermal decomposition, including loss of physically adsorbed solvents, removal of coordinated or lattice water molecules, degradation of organic ligand frameworks, and formation of final inorganic residues. In Schiff base compounds, TGA analysis is frequently used to evaluate decomposition temperatures and compare the thermal stability of free ligands with their corresponding metal complexes [16].

In many cases, metal complexes of Schiff base ligands have been reported to exhibit higher thermal stability than the corresponding free ligands, which is commonly attributed to stronger metal–ligand interactions and increased structural rigidity after coordination [17]. However, this behavior is not universal. The thermal stability of Schiff base complexes depends on several factors, including the nature of the metal ion, the coordination geometry, the presence of lattice or coordinated water molecules, and the ligand framework itself. For instance, some lanthanide complexes decompose at lower temperatures than their parent ligands due to the presence of weakly bound solvent molecules [18], while certain polymeric or highly conjugated free ligands may exhibit thermal stability comparable to that of their complexes [19]. Consequently, thermal stability should be evaluated on a case-by-case basis rather than assumed as a general trend.

DSC provides complementary information by measuring heat flow associated with thermal events such as melting, crystallization, glass transition, and phase transformations. DSC analysis is useful for understanding the physical properties of Schiff base derivatives and evaluating their stability during heating processes. The combination of DSC and TGA allows researchers to obtain a more complete thermal profile of synthesized compounds and identify important transitions occurring within the material [20].

Thermal analysis is particularly significant for Schiff base compounds designed for practical applications, including catalysts, advanced materials, sensors, and pharmaceutical formulations. Knowledge of thermal stability helps predict material performance under

operating conditions and supports the selection of appropriate compounds for specific technological applications. Although thermal analysis does not directly determine molecular structure, its combination with spectroscopic and crystallographic techniques provides a comprehensive evaluation of newly synthesized Schiff base ligands and their complexes [21].

Table 3: Common characterization techniques used for Schiff base ligands [21]

Characterization techniques	Main information obtained
FTIR spectroscopy	Identification of functional groups and confirmation of C=N bond formation
NMR spectroscopy	Molecular structure, proton and carbon environments
UV-Vis spectroscopy	Electronic transitions and conjugation effects
Mass spectroscopy	Molecular mass and fragmentation patterns
XRD-Xray diffraction	Crystal structure and molecular arrangement
TGA/DSC	Thermal stability and decomposition behavior

Critical assessment: TGA and DSC are essential for evaluating thermal stability, decomposition pathways, and phase transitions. TGA can reveal the loss of solvent molecules, coordinated water, and organic framework degradation, while DSC identifies melting, crystallization, and glass transitions. However, these techniques (Table 4) do not provide direct structural information; decomposition steps must be interpreted in conjunction with known molecular formulas and spectroscopic data. Additionally, TGA/DSC results can be influenced by heating rate, sample mass, and atmosphere, requiring careful standardization for reproducibility. Therefore, thermal analysis is best regarded as a property-evaluation tool that complements structural techniques.

Table 4: Critical comparison of characterization techniques for Schiff base ligands

Technique	Information Reliably Obtained	Limitations	Requires Complementary Analysis
FTIR	Presence of functional groups (C=N, O–H, N–H, and C–O); coordination-induced shifts	Overlapping bands; no connectivity or stereochemistry; C=N band alone not definitive proof	NMR, MS, and XRD
^{1}H-/^{13}C-NMR	Chemical environment, connectivity, symmetry, tautomerism, and substituent effects	Requires solubility; limited for paramagnetic complexes; and no direct C=N proof	FTIR, MS, and XRD
UV-Vis	Electronic transitions (π→π, $n\rightarrow\pi$), conjugation, MLCT, and photophysical properties	Low structural resolution; broad overlapping bands; and solvent/pH sensitive	NMR, MS, and XRD
MS	Molecular mass, molecular formula, fragmentation pathways, and purity	Cannot distinguish isomers; thermal lability issues; and complex fragmentation	NMR, FTIR, and XRD
XRD	3D molecular structure, bond lengths/angles, coordination geometry, and crystal packing	Requires single crystals; not suitable for amorphous samples	NMR, FTIR, and MS
TGA/DSC	Thermal stability, decomposition steps, phase transitions, and solvent/water loss	No direct structural information; heating-rate dependent	FTIR, NMR, and XRD

Applications of Schiff Base Ligands

The remarkable structural flexibility, electron-donating ability, and coordination behavior of Schiff base ligands have resulted in their extensive applications in various scientific fields. The ability to modify their molecular structures through different combinations of aldehydes, ketones, and amines allows researchers to design compounds with specific chemical, biological, and functional properties. Schiff base ligands and their metal complexes have been widely investigated in medicinal chemistry, catalysis, materials science, environmental applications, and analytical chemistry [25].

Biological Applications

Schiff base ligands and their metal complexes have attracted considerable attention in biological research due to their diverse pharmacological activities. The presence of the azomethine functional group and additional donor atoms contributes to their ability to interact with biological targets, including proteins, enzymes, and nucleic acids. Numerous studies have reported antibacterial, antifungal, antioxidant, antiviral, anti-inflammatory, and anticancer activities for both free Schiff base ligands and their corresponding metal complexes [26].

Antimicrobial activity is one of the most extensively investigated properties of Schiff base compounds. Many Schiff base derivatives demonstrate significant inhibitory effects against a wide range of bacterial strains, including both Gram-positive and Gram-negative microorganisms. In many reported cases, coordination of Schiff base ligands to metal ions has been observed to enhance antimicrobial activity relative to the free ligands. This enhancement has been attributed to factors such as increased lipophilicity, improved membrane penetration, and, in some systems, the generation of reactive oxygen species [27]. However, this trend is not universal and depends strongly on the nature of the metal ion, the ligand structure, the coordination geometry, and the bacterial strain tested. For example, while Cu(II) and Zn(II) complexes frequently show improved activity, some Ni(II) and Co(III) complexes have been reported to exhibit comparable or even reduced activity relative to their parent ligands [10,17]. Therefore, generalizations should be made with caution, and each system requires individual evaluation. Metal complexes containing copper, nickel, cobalt, zinc, and other transition metals have been reported in numerous studies to show improved antimicrobial performance compared with their uncoordinated ligands [28]. However, the magnitude of enhancement varies considerably with the metal ion, ligand structure, and microbial strain. In some cases, complexation has resulted in no significant improvement or even reduced activity, highlighting the need for systematic structure–activity relationship studies [29].

The anticancer potential of Schiff base ligands and their metal complexes has also received significant research interest. Several synthesized compounds have demonstrated cytotoxic effects against different cancer cell lines through mechanisms involving DNA interaction, oxidative stress induction, enzyme inhibition, and apoptosis activation. Transition metal complexes are particularly promising in this area because metal centers can participate in biological redox processes and enhance the therapeutic activity of the organic ligand [30].

In addition to antimicrobial and anticancer properties, Schiff base compounds have exhibited antioxidant and anti-inflammatory activities. The presence of phenolic hydroxyl groups, aromatic systems, and heteroatom-containing functional groups can contribute to radical scavenging ability and biological activity. Structural modification of Schiff base ligands provides opportunities to optimize their pharmacological properties and improve their selectivity toward specific biological targets [31].

Although many Schiff base derivatives show promising biological activities, further investigations are required to evaluate their toxicity, pharmacokinetic behavior, mechanism of action, and potential clinical applications. Therefore, Schiff base chemistry continues to represent an important platform for the development of new bioactive molecules and metal-based therapeutic agents [32].

Representative studies illustrate the quantitative biological performance of Schiff base ligands and their complexes (Table 5). For example, Cu(II) complexes of salicylaldehyde-derived Schiff bases have shown MIC values of 2–8 µg/mL against *S. aureus* and *E. coli*, significantly lower than those of the free ligands (32–64 µg/mL). Schiff base–Cu(II) and –Zn(II) complexes derived from 2-hydroxy-1-naphthaldehyde exhibited IC_{50} values of 5–15 µM against MCF-7 and HeLa cell lines, compared with 40–80 µM for the uncoordinated ligands. Antioxidant studies using the DPPH assay reported IC_{50} values of 10–30 µM for phenolic Schiff bases, approaching that of ascorbic acid ($IC_{50} \approx 8$ µM).

Table 5: Representative biological applications of Schiff base ligands and their metal complexes

Compound / Complex	Application	Key Performance Parameter	Advantages	Limitations	Ref.
Salicylaldehyde–Cu(II)	Antibacterial	MIC = 2–8 µg/mL (*S. aureus*, *E. coli*)	Enhanced lipophilicity; broad-spectrum	Cytotoxicity not fully evaluated	[10]
2-Hydroxy-1-naphthaldehyde–Zn(II)	Anticancer	IC_{50} = 5–15 µM (MCF-7, HeLa)	High selectivity; apoptosis induction	*In vivo* data lacking	[11]
Phenolic Schiff base	Antioxidant	IC_{50} = 10–30 µM (DPPH)	Radical scavenging; low cost	Stability issues in aqueous media	[12]
Schiff base–Co(III)	Antiviral	EC_{50} = 8–20 µM	DNA binding; enzyme inhibition	Mechanism not fully elucidated	[13]
Schiff base–Ni(II)	Anti-inflammatory	IC_{50} = 12–25 µM	COX-2 inhibition	Limited pharmacokinetic data	[14]

It is important to emphasize that the biological, catalytic, and thermal properties of Schiff base ligands and their metal complexes are highly system-dependent. General trends should therefore be interpreted with caution, and conclusions should be drawn only after considering the specific ligand structure, metal ion, coordination environment, and experimental conditions. This underscores the need for systematic structure–property relationship studies in future research.

Catalytic Applications

Schiff base ligands and their metal complexes have become important components in the field of catalysis due to their excellent coordination ability, structural diversity, and tunable electronic properties. The presence of donor atoms such as nitrogen, oxygen, and sulfur enables Schiff base ligands to coordinate and stabilize a wide range of metal centers, and in many systems, this coordination environment has been shown to influence catalytic activity and selectivity [33–35]. However, the relationship between ligand structure and catalytic performance is complex and depends on factors such as the metal ion, oxidation state, steric environment, and reaction conditions. Therefore, the catalytic behavior of Schiff base complexes cannot be generalized solely on the basis of donor atom availability.

One of the most significant applications of Schiff base complexes is their use as oxidation catalysts. Metal complexes containing Schiff base ligands, particularly those of transition metals such as manganese, copper, cobalt, nickel, and iron, have demonstrated high catalytic activity in the oxidation of organic substrates. These catalysts can promote oxidation reactions under relatively mild conditions while providing improved selectivity and efficiency. The electronic environment created by the Schiff base ligand plays an important role in controlling the redox

properties of the metal center and enhancing catalytic performance [21].

Schiff base metal complexes are also widely applied in carbon–carbon and carbon–heteroatom bond-forming reactions. Their ability to activate substrates and stabilize reactive intermediates makes them valuable catalysts for processes such as coupling reactions, condensation reactions, and cyclization reactions. By modifying the ligand structure, researchers can adjust steric and electronic factors around the active metal center, allowing optimization of catalytic activity and selectivity [22]. In addition to homogeneous catalysis, Schiff base compounds have been incorporated into heterogeneous catalytic systems and advanced functional materials. Immobilization of Schiff base complexes on solid supports, nanoparticles, polymers, or porous materials can improve catalyst stability, recyclability, and separation efficiency. These developments have increased the potential applications of Schiff base-based catalysts in industrially relevant chemical processes [23].

Despite their considerable advantages, challenges remain regarding catalyst recovery, long-term stability, and large-scale industrial implementation. Future research is focused on designing environmentally friendly Schiff base catalysts with higher efficiency, improved recyclability, and reduced environmental impact. The combination of green chemistry principles with advanced ligand design is expected to further expand the role of Schiff base compounds in modern catalytic systems [24].

Quantitative catalytic performance data demonstrate the efficiency of Schiff base complexes. Mn(III)–salen complexes catalyze the epoxidation of styrene with conversions of 85–98% and selectivities above 90% within 2–4 h. Cu(II)–Schiff base complexes have been employed in the oxidative coupling of phenols, achieving TOF values of 150–300 h^{-1}. Pd(II)–Schiff base complexes show high activity in Suzuki–Miyaura coupling reactions, with yields of 90–99% and TON values up to 9,500. Immobilized Schiff base catalysts on silica or magnetic nanoparticles maintained >90% activity after five consecutive cycles (Table 6).

Table 6: Representative catalytic applications of Schiff base metal complexes

Catalyst	Reaction	Key Performance Parameter	Advantages	Limitations	Ref.
Mn(III)–salen	Styrene epoxidation	Conversion 85–98%; selectivity >90%	Mild conditions; high selectivity	Sensitive to oxidant	[31]
Cu(II)–Schiff base	Phenol oxidative coupling	TOF = 150–300 h^{-1}	Earth-abundant metal	Limited substrate scope	[32]
Pd(II)–Schiff base	Suzuki–Miyaura coupling	Yield 90–99%; TON up to 9,500	High activity; broad substrate scope	Cost of Pd; recovery issues	[33]
Fe(III)–Schiff base	Oxidation of alcohols	Conversion 80–95%	Green oxidant (H_2O_2)	Deactivation over time	[34]
Immobilized Schiff base–Cu(II)	Click chemistry	>90% activity after 5 cycles	Recyclability; easy separation	Mass transfer limitations	[35]

Environmental and Material Applications

The unique structural and electronic properties of Schiff base ligands have expanded their applications beyond biological and catalytic fields into environmental science and advanced materials. Their ability to coordinate with various metal ions, bind specific molecules, and exhibit tunable optical and electronic properties has made them valuable candidates for environmental remediation, chemical sensing, and functional material development [4].

One of the important environmental applications of Schiff base compounds is their use in the removal and detection of heavy metal ions. Schiff base ligands containing donor atoms

such as nitrogen, oxygen, and sulfur can interact strongly with toxic metal ions through coordination interactions. These properties enable their application as selective adsorbents or sensing materials for detecting pollutants such as mercury, lead, copper, and other hazardous metal species in aqueous environments. The design of Schiff base-based sensors with improved selectivity and sensitivity has become an active area of research [5].

Schiff base derivatives have also attracted significant attention in the development of fluorescent and colorimetric sensors. Their conjugated structures and ability to undergo electronic changes upon interaction with analyte molecules or metal ions allow them to generate measurable optical responses. These characteristics make Schiff base compounds suitable for applications in chemical detection, environmental monitoring, and analytical devices. Structural modification of the ligand framework can further improve sensing performance by controlling selectivity, response time, and detection limits [6].

In materials science, Schiff base ligands have been utilized in the preparation of coordination polymers, metal–organic frameworks (MOFs), nanoparticles, and other functional materials. Their multifunctional donor sites allow the construction of organized structures with interesting magnetic, electrical, catalytic, and optical properties. Such materials have potential applications in energy storage, molecular electronics, photocatalysis, and advanced sensing technologies [7].

Furthermore, Schiff base-based materials have been investigated for corrosion inhibition due to their ability to adsorb onto metal surfaces and form protective layers. The presence of heteroatoms and aromatic systems enhances interaction with metal substrates, reducing corrosion processes in aggressive environments. These findings demonstrate the broad technological importance of Schiff base chemistry and highlight its continuing contribution to the development of innovative functional materials [8].

Table 7 shows that overall, the versatility of Schiff base ligands allows their application across multiple scientific disciplines. Their combination of structural adaptability, coordination ability, and tunable properties continues to drive research toward new environmental technologies and advanced materials with improved performance and sustainability [9].

Table 7: Applications of Schiff base ligands and their metal complexes

Application area	Main role of Schiff base ligands
Biological applications	Inflammatory-antimicrobial, anticancer, and antioxidant activities
Catalytic applications	Oxidation, reduction, and organic transformation reactions
Environmental applications	Detection and removal of heavy metal ions
Chemical sensors	Fluorescent and colorimetric sensing of analytes
Materials science	Preparation of coordination polymers, MOFs, and functional materials

Schiff base ligands have been widely applied in environmental remediation, particularly for heavy metal detection and removal (Table 8). Colorimetric and fluorescent sensors based on Schiff bases have achieved detection limits as low as 0.05–0.5 μM for Cu^{2+}, Hg^{2+}, and Pb^{2+}. Adsorption studies using Schiff base-functionalized materials reported the maximum adsorption capacities (q_{max}) of 120–350 mg/g for Pb(II) and 80–200 mg/g for Cd(II). Schiff base-modified electrodes have been used for electrochemical detection of heavy metals with detection limits in the nanomolar range.

Table 8: Representative environmental applications of Schiff base ligands and materials

Material / Sensor	Application	Key Performance Parameter	Advantages	Limitations	Ref.
Schiff base fluorescent sensor	Cu^{2+} detection	LOD = 0.05–0.5 μM	High selectivity; rapid response	Interference from Fe^{3+}	[3]
Schiff base-functionalized silica	Pb(II) adsorption	q_{max} = 120–350 mg/g	High capacity; reusable	pH-dependent	[5]
Schiff base-modified electrode	Hg^{2+} detection	LOD = 1–10 nM	High sensitivity	Electrode fouling	[7]
Schiff base–chitosan composite	Cd(II) removal	q_{max} = 80–200 mg/g	Biodegradable; low cost	Slow kinetics	[8]
Schiff base colorimetric sensor	Pb^{2+} detection	Visual detection at 1 μM	Naked-eye detection	Semi-quantitative	[9]

Electrochemical and Bioanalytical Applications

Schiff base ligands and their metal complexes have also shown promising potential in electrochemical and bioanalytical applications. The redox-active nature of many transition metal Schiff base complexes enables their use as electrode modifiers, electrochemical sensors, and electron-transfer mediators. Modified electrodes incorporating Schiff base complexes have been employed for the sensitive detection of biologically important analytes such as dopamine, ascorbic acid, glucose, and hydrogen peroxide. Furthermore, the ability of Schiff base ligands to coordinate with metal ions has been exploited in the design of electrochemical biosensors for heavy metal detection and environmental monitoring. In bioanalytical chemistry, Schiff base-based fluorescent and electrochemical probes have been developed for the selective recognition of biomolecules, including amino acids, proteins, and DNA. Although these applications are still emerging compared with the well-established biological and catalytic uses of Schiff bases, they represent a rapidly growing area of research with significant potential for future development.

Future Perspectives and Challenges

Despite significant advances in the synthesis, characterization, and application of Schiff base ligands, several challenges remain that require further investigation. The continuous development of new Schiff base structures with improved properties depends on the integration of advanced synthetic strategies, modern characterization techniques, and computational approaches. Future research is expected to focus on designing multifunctional Schiff base ligands with enhanced biological activity, catalytic efficiency, and environmental performance [6]. One of the major challenges in Schiff base chemistry is the development of sustainable synthetic procedures that minimize environmental impact while maintaining high reaction efficiency. Although green synthesis, solvent-free methods, and microwave-assisted approaches have shown promising results, further improvements are needed to achieve scalable, economical, and industrially applicable processes. The use of renewable starting materials, recyclable catalysts, and environmentally friendly solvents will likely become increasingly important in future research [7]. In the field of medicinal chemistry, Schiff base compounds continue to show promising biological activities; however, detailed investigations regarding toxicity, pharmacokinetics, molecular mechanisms, and clinical applicability are still limited. Future studies should combine experimental approaches with computational modeling, molecular docking, and biological evaluation to develop safer and more effective therapeutic candidates [8]. For catalytic and material applications, future research should focus on improving catalyst stability, selectivity, recyclability, and performance under practical operating conditions. The incorporation of Schiff base ligands into advanced materials such as metal–organic frameworks, nanostructured systems, and hybrid materials may provide new

opportunities for applications in energy, sensing, and environmental technologies [9].
Overall, the future of Schiff base chemistry lies in the rational design of multifunctional ligands through the combination of synthetic chemistry, computational methods, and advanced analytical techniques. Continued interdisciplinary research will enable the development of innovative Schiff base-based systems capable of addressing challenges in medicine, catalysis, environmental protection, and materials science [10].

In materials science, Schiff base ligands have been used to construct coordination polymers, MOFs, and functional nanomaterials (Table 9). Schiff base-based MOFs have shown surface areas of 500–2,500 m^2/g and pore sizes of 0.5–2 nm, making them suitable for gas storage and separation. Schiff base–metal complexes incorporated into polymer matrices exhibited tunable optical band gaps (2.0–3.5 eV) and were used in organic light-emitting diodes (OLEDs) and photovoltaic devices. Schiff base-based corrosion inhibitors reduced corrosion rates by 85–98% at concentrations of 50–200 ppm.

Table 9: Representative materials science applications of Schiff base ligands

Material	Application	Key Performance Parameter	Advantages	Limitations	Ref.
Schiff base MOF	Gas storage/separation	Surface area = 500–2,500 m^2/g	High porosity; tunable pores	Moisture sensitivity	[2]
Schiff base–polymer composite	OLED/optoelectronics	Band gap = 2.0–3.5 eV	Tunable emission	Stability issues	[8]
Schiff base corrosion inhibitor	Steel protection	Inhibition efficiency = 85–98%	Low cost; easy application	Toxicity concerns	[9]
Schiff base–metal complex	Magnetic materials	Saturation magnetization = 10–40 emu/g	Tunable magnetic properties	Aggregation	[11]
Schiff base–nanoparticle hybrid	Catalysis/sensing	Particle size = 5–50 nm	High surface area	Synthesis complexity	[12]

Recent research on Schiff base ligands has focused on developing more sustainable synthetic methods, improving their functional properties, and expanding their applications in advanced fields. Green synthesis approaches, including solvent-free reactions and environmentally friendly reaction conditions, have attracted considerable attention due to their reduced environmental impact [6,7].
Additionally, the combination of Schiff base ligands with transition metals has provided new opportunities in biological, catalytic, and material applications. Advanced characterization techniques and computational studies have improved the understanding of their structural properties and reactivity [8-10].
Despite significant progress, several challenges remain, including the development of highly selective and stable compounds, investigation of toxicity and environmental effects, and improvement of large-scale production methods. Future studies should focus on designing multifunctional Schiff base systems with enhanced performance for practical applications [11,12].

Conclusion

Schiff base ligands continue to attract significant research interest due to their straightforward synthesis, structural tunability, and broad application potential. This review has critically examined recent advances in their synthesis, characterization, and applications, with particular emphasis on comparative analysis of synthetic methodologies and the complementary roles of spectroscopic, crystallographic, and thermal techniques. This review has summarized recent developments in the synthesis and characterization of new Schiff

base ligands, including conventional methods, green synthetic approaches, microwave-assisted techniques, and solvent-free strategies. Modern characterization methods such as FTIR, NMR, UV–Vis spectroscopy, MS, XRD, and thermal analysis have been highlighted as essential tools for confirming molecular structures and understanding the physicochemical properties of synthesized compounds. Overall, the continuous development of new Schiff base ligands confirms their importance as multifunctional chemical platforms with promising contributions to modern chemistry, biotechnology, and materials science. Emerging electrochemical and bioanalytical applications of Schiff base ligands, including their use in sensors and biosensors, further demonstrate the versatility of these compounds and open new avenues for future research.

Acknowledgements

The authors would like to thank the University of Torbat-e Jam, Iran for supporting this work.

ORCID

Milad Mahjoobizadeh
http://orcid.org/0009-0006-6068-3327
Vahidreza Darugar
http://orcid.org/0000-0003-4901-9389

References

[1] Moanta, A. Classification, synthesis, isomerism, and spectral characterization of schiff bases. *Mini-Reviews in Organic Chemistry*, **2025**, 22, 718-737.
[2] Marinova, P.E., Tamahkyarova, K.D. Synthesis, investigation, biological evaluation, and application of coordination compounds with Schiff base—A review. *Compounds*, **2025**, 5, 14.
[3] Malav, R., Ray, S. Recent advances in the synthesis and versatile applications of transition metal complexes featuring Schiff base ligands. *RSC Advances*, **2025**, 15, 22889-22914.
[4] Uddin, E., Sardar, M.N., Reza, M.S., Hasan, M.S., Talukder, M.T., Hoque, M.M., Paul, P., Khatun, M.S., Hossen, M.F., Asraf, M.A., Kudrat-E-Zahan, M. Emerging pharmaceutically active drugs: synthesis and pharmacology of Schiff base ligands with their metal complexes. *Discover Chemistry*, **2025**, 2, 153.
[5] Schiff, H. Mittheilungen aus dem Universitätslaboratorium in Pisa: Eine neue Reihe organischer Basen. *Justus Liebigs Annalen der Chemie*, **1864**, 131, 118-119.
[6] Przybylski, P., Huczynski, A., Pyta, K., Brzezinski, B., Bartl, F. Biological properties of schiff bases and azo derivatives of phenols. *Current Organic Chemistry*, **2009**, 13, 124-148.
[7] da Silva, C.M., da Silva, D.L., Modolo, L.V., Alves, R.B., de Resende, M.A., Martins, C.V.B., de Fátima, Â. Schiff bases: A short review of their antimicrobial activities. *Journal of Advanced Research*, **2011**, 2, 1-8.
[8] Abu-Dief, A.M., Mohamed, I.M.A. A review on versatile applications of transition metal complexes incorporating Schiff bases. *Beni-Suef University Journal of Basic and Applied Sciences*, **2015**, 4, 119-133.
[9] Dhar, D.N., Taploo, C. Schiff bases and their applications. *Journal of Scientific and Industrial Research*, **1982**, 41(8), 501-506.
[10] Singh, K., Barwa, M., Tyagi, P. Synthesis, characterization and biological studies of Co (II), Ni (II), Cu (II) and Zn (II) complexes with bidentate Schiff bases derived by heterocyclic ketone. *European Journal of Medicinal Chemistry*, **2006**, 41, 147-153.
[11] Kaes, C., Katz, A., Hosseini, M.W. Bipyridine: the most widely used ligand. A review of molecules comprising at least two 2,2'-bipyridine units. *Chemical Reviews*, **2000**, 100, 3553-3590.
[12] Luo, H., Liu, S., Rettig, S.J., Orvig, C. Rhenium and technetium complexes from pentadentate (N_3O_2) and tetradentate (N_2O_2) Schiff base ligands. *Canadian Journal of Chemistry*, **1995**, 73, 2272-2281.
[13] Patai, S. The Chemistry of the carbon-nitrogen double bond. *Interscience Publishers*, **1970**.
[14] Hameed, A., al-Rashida, M., Uroos, M., Abid Ali, S., Khan, K.M. Schiff bases in medicinal chemistry: A patent review (2010-2015). *Expert Opinion on Therapeutic Patents*, **2016**, 27, 63-79.
[15] Kajal, A., Bala, S., Kamboj, S., Sharma, N., Saini, V. Schiff Bases: A versatile pharmacophore. *Journal of Catalysts*, **2013**, 2013, 1-14.
[16] Mallikarjun, G. Synthesis, characterization and biological activity of Schiff base metal complexes. *Asian Journal of Research in Chemistry*, **2017**, 10, 587.
[17] Chandra, S., Kumar, U. Spectral and magnetic studies on manganese (II), cobalt (II) and nickel (II) complexes with Schiff bases. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, **2005**, 61, 219-224.
[18] Warner, J.C. Green chemistry: Theory and practice. *Abstracts of Papers of The American Chemical Society*. **2012**, 244, 19-23.
[19] Sheldon, R.A. Green solvents for sustainable organic synthesis: state of the art. *Green Chemistry*, **2005**, 7, 267.

[20] Kappe, C.O. Controlled microwave heating in modern organic synthesis. *Angewandte Chemie International Edition*, **2004**, 43, 6250-6284.
[21] Leadbeater, N.E. Microwave heating as a tool for sustainable chemistry, **2010**, CRC press.
[22] Varma, R.S. "Greener" chemical syntheses using mechanochemical mixing or microwave and ultrasound irradiation. *Green Chemistry Letters and Reviews*, **2007**, 1, 37-45.
[23] Rostami-Vartooni, A., Kia, R., Abbasi, F., Kiamehr, M., Raithby, P.R. Structural and computational studies of new Schiff base ligands derived from 2-(methylamino)- and 2-(ethylamino) benzaldehyde: synthesis, characterization, QTAIM, NCI plot, TD-DFT and energy decomposition analysis. *Acta Crystallographica Section B Structural Science, Crystal Engineering and Materials*, **2025**, 81, 540-549.
[24] Abid, M.A., Al-Jumaili, M.A.J., Abed, A.A. Synthesis, characterization and gene expression of novel bis-Schiff base compounds and study their anti-bacterial activity using different genes. *Results in Chemistry*, **2025**, 13, 102042.
[25] Takhar, P., Chhapoliya, M., Janmeda, P., Singh, D. Advances in coordination complexes of Schiff bases with diverse nuclearities and functionalities: Molecular properties, supramolecular aspects, and applications. *Coordination Chemistry Reviews*, **2026**, 564, 218001.
[26] Silverstein, R.M., Bassler, G.C. Spectrometric identification of organic compounds. *Journal of Chemical Education*, **1962**, 39(11), 546.
[27] Nakamoto, K. Infrared and Raman spectra of inorganic and coordination compounds, part B: applications in coordination, organometallic, and bioinorganic chemistry. *John Wiley & Sons*, **2009**, 1-70.
[28] Pavia, D., Lampman, G.M., Kriz, G.S., Vyvyan, J. Introduction to spectroscopy, Fifth edit. *Cengage Learning, Stamford, CT*, **2015**, 784.
[29] Hughes, E., Wang, X.X., Sabol, L., Barton, K., Hegde, S., Myakala, K., Krawczyk, E., Rosenberg, A., Levi, M. Role of nuclear receptors, lipid metabolism, and mitochondrial function in the pathogenesis of diabetic kidney disease. *American Journal of Physiology-Renal Physiology*, **2025**, 329, F510-F547.
[30] Cozzi, P.G. Metal–Salen Schiff base complexes in catalysis: practical aspects. *Chem Soc Rev*, **2004**, 33, 410-421.
[31] Gupta, K.C., Sutar, A.K. Catalytic activities of Schiff base transition metal complexes. *Coordination Chemistry Reviews*, **2007**, 252, 1420-1450.
[32] Al Zoubi, W., Ko, Y.G. Schiff base complexes and their versatile applications as catalysts in oxidation of organic compounds: part I. *Applied Organometallic Chemistry*, **2017** , 31(3), e3574.
[33] Chowdhury, J., Mondal, A., Chattopadhyay, S. Binuclear Ruthenium (II) Organometallics Incorporating the Bridging Schiff Base Ligands: Synthesis, Spectroscopic, Structural and Theoretical Studies. *ES Chemistry and Sustainability*, **2026**, 7, 2063.
[34] Qin, W., Long, S., Panunzio, M., Biondi, S. Schiff bases: A short survey on an evergreen chemistry tool. *Molecules*, **2013**, 18, 12264-12289.
[35] Dhanaraj, C.J., Raj, S. Synthesis, characterization and biological studies of Schiff base metal complexes derived from 4-aminoantipyrine, acetamide and p-phenylenediamine. *Inorganic Chemistry Communications*, **2020**, 119, 108087.

Copyright © 2026 Sami Publishing Company. This article is licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0).